



Study of the Efficacy of Porous Carbons Using Modern Methods

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Abstract: The purpose of this work was to study purolate (porous carbons from the Kuzbass deposits, Russia). Thermograms in the temperature range 150–700 °C showed an up to 8.7% mass loss in the purolate samples. It was proven that purolate has a large range of particle size (from 0.1 to 3 mm) and pH (8.0–9.0) and a low total pore volume in water (0.5 cm³/g). It was found that in addition to C and O₂, Zn (5,346.8 mg/kg), Ba (256 mg/kg), Sr (304 mg/kg), Cu (541 mg/kg), and MnO (119 mg/kg) are present in significant amounts in purolate; it does not contain Al₂O₃, SiO₂, Rb, and Zr. It was established that the service life of the sorbent layer is 380 min at an adsorption temperature of 28–30 °C (analysis of the adsorption breakthrough curve). The final degree of purification from the model mixture ranged from 35.4% for manganese ions to 98.1% for iron ions. Analysis of the kinetic curves of ion extraction found that the highest adsorption (0.07 g/g) for 250 min was observed during the extraction of manganese ions, the lowest (0.045 g/g) for 300 min, during the extraction of nitrite ions. The development of a new technology using anthracite-based adsorbents for treating water from coal mining operations would help address environmental concerns in resource-dependent areas and contribute to the rehabilitation and revitalization of aquatic ecosystems.

Keywords: Activated carbon; Adsorption isotherms; Purification; Sorbent; X-ray analysis

1 Introduction

Adsorption is the accumulation of a liquid or gas phase on the surface or wall of a solid. This phenomenon is characterized by the interaction of the volume phase of the adsorbent (adsorptive) and the solid material (adsorbent) [1]. The primary experimental technique for nitrogen porometry involves measuring the gas absorption temperature at atmospheric pressure and a temperature close to the welding temperature.

The kinetic parameters of the extraction process allow tracing the purification dynamics by sorbent, that is, the speed of the extraction process [2]. The process of dispersing extracted components within an independent particle determines the extraction mechanism and allows predicting the extraction rate under dynamic conditions [1]. At the initial point of contact between the extracted component and the sorption material, at the earliest stages of interaction, it is possible to discuss the mechanisms and, over an extended period, observe changes in the behavior of the components [3].

In the case of porous adsorbent materials, in addition to external mass transfer, internal mass transfer also plays a significant role. This refers to the movement of the adsorbate within the pores of the adsorbent, driven by a concentration gradient. The mechanism of such transfer may be dependent on the concentration of the adsorption agent and the size of the pores [4, 5]. These dependencies allow describing the structure model of the granules of materials, since to study the kinetics of adsorption, it is essential to determine which model (quasi-homogeneous or porous) best describes the sorbent grains.

The kinetic parameters of the sorption process are affected by various factors, such as the specific surface area, the structure of carbon-based sorbents, and conditions for the transfer of molecules of the substance being adsorbed onto

the surface of the adsorbent grain.

Considering the complexity of pollutant sorption from aqueous solutions to identify the rate-limiting step, the suitability of various kinetic models has been evaluated. These models can help characterize the sorption process and determine the limiting step of extraction (rational conditions for implementing the water purification process). The adsorption mechanisms may be provided by the kinetic models, which determine the likelihood of a particular adsorption process [6]. This data is also essential for understanding the dynamics of the process and designing an effective adsorption system.

Porous carbon materials (PCM) are widely used in various technological processes for separating liquid and gaseous media due to their high sorption capacity. PCMs are increasingly used as catalysts and hemosorbents, chromatography and natural gas storage systems, as well as carriers of carbon-carbon compounds [7]. PCMs have the following unique properties: high oxidative and catalytic activity, good stability in non-oxidizing media, specific surface area from 0.1 to 10^3 m²/g, and pore size from microns to hundreds of microns. The forms of the final products include powders, particles, block products, films, fibers, and fibrous materials [8].

Currently, the global production of materials containing carbon is approximately 1 million tons annually [7, 9]. The increase in production and expansion of the range of PCMs are achieved by using various carbonaceous raw materials, such as wood [10], coal and brown coal [11], agricultural waste [12], cakes [8] and polymers used in the production of materials [13], liquid and gaseous hydrocarbons [14], and carbonaceous industrial and household waste.

Physical or chemical activation methods are often used to produce PCMs for use as adsorbents or catalysts [7, 8]. The physical manufacture of PCMs includes the following stages: preparation of raw materials (sorting, grinding, drying, etc.), pyrolysis (heat treatment without access to an oxidizer at a temperature of 550–1,000 °C); activation (heat treatment in the presence of an oxidizer, CO₂, or water vapor at 700–1,000 °C).

The production of PCMs by thermochemical activation is based on the introduction of chemical additives into the raw materials, which are then carbonized in an inert atmosphere or in the presence of gaseous oxidants. The conversion of raw materials into PCMs occurs under the action of acid-based or redox catalysts (oxidation of ZnCl₂, Al₂O₃, H₃PO₄, carbonates, or alkali metals) [7, 8, 15]. By simultaneous carbonization and activation, usually at temperatures below 700 °C, catalysts can modify aliphatic fragments by removing oxygen, hydrogen, and other foreign substances. The result is a PCM with an improved porous structure. To use PCMs effectively, it is necessary to understand their structure and structural properties.

The main challenge limiting the use of anthracite in the production of carbon adsorbents is its relatively low reactivity. Thus, traditional methods of steam or CO₂ activation are not effective enough. The developed industrial methods of anthracite activation have low yields, and it is impossible to obtain carbon adsorbents of a sufficiently wide spectrum of action on their basis [16]. Traditional methods of physical activity typically require a significant amount of time and involve high temperatures to produce carbon adsorbents with good adsorption properties. At the same time, more than 50% of the initial mass of anthracite is lost during combustion.

There is evidence that low-temperature chemical reactions occurring in anthracite (such as HClO₄ and HNO₃) increase reactivity during subsequent physical exertion [17].

The initial chemical modification of anthracite with nitric and sulfuric acids significantly changes the composition and structure of anthracite. By changing the synthesis conditions, activated anthracite with a surface area of up to 1,000 m²/g, low ash content, and controlled pore size distribution were obtained. We found that the conditions of anthracite treatment with a mixture of sulfuric and nitric acids ensure complete dissolution and formation of a new carbon phase during precipitation from the solution. Short-term heat treatment (several minutes) of carbon materials at a temperature of 900 °C makes it possible to obtain a PCM with a reinforced porous structure.

PCMs produced by chemical activation of anthracite coal with mineral acids have a porous microstructure. The pore volume is from 0.133 to 0.253 cm³/g. The size of the micropores varied from 0.7 to 2.5 nm, with an average size of 0.94 nm [12].

Thermally treated and modified anthracite has a high iodine absorption capacity of 71%. Please note that for most industrial applications, this value is approximately 60%. The yield of PCMs from anthracite coals modified with mineral acid is up to 84% of the initial mass. When using conventional steam activation of anthracite, from 30% to 50% of the starting material is lost [18].

To develop a new method for the synthesis of nanoporous carbon materials, the activation process of anthracite coals from several Russian deposits containing lithium hydroxides, sodium hydroxides and potassium hydroxides were studied. Studies have shown that the resulting carbon material experimentally depends on the properties of anthracite and activator, as well as the conditions of the activation process.

When studying the sorption properties of anthracite using alkalis, in the series LiOH < NaOH < KOH, the activation potential of hydroxide increases. By selecting the system parameters (temperature, reactor ratio, reaction medium) for the chemical activation of anthracite with K, Li, and Na hydroxides, it is possible to obtain nanocarbon materials with a particularly high surface area [4].

The most significant change in the surface area of the carbon material during the activation of anthracite occurred

in the presence of potassium hydroxide. The surface area of this material was approximately 3,000 m²/g, and the pore volume was approximately 1.77 cm³/g. The average pore size of the material was 2.2 nm. When anthracite was activated in the presence of sodium hydroxide, the resulting carbon material has a specific surface area of 1,500 m²/g. In contrast, the lithium hydroxide/anthracite system has a specific surface area of 240 m²/g and average pore sizes of 2.6 and 1.0 nm, respectively [5].

When studying carbon porous sorbents, Zhang et al. [19] investigated the effect of steam activation on the evolution of the porous structure and surface chemistry of activated carbon (AC) obtained from bamboo waste. The study showed that during steam treatment, additional micropores are created on the sorbent surface, and existing ones are expanded. This increases the sorbent surface area and the total volume of micropores. The sorbent was activated by steam at a temperature of 850°C for 120 min. AC was obtained with an average specific surface area of 1,210 m²/g and a total porosity of 0.542 cm³/g. The results of Fourier transform infrared spectroscopy and X-ray photoelectron spectroscopy indicate a substantial number of active functional groups on the surface of AC. The adsorption of steam-treated AC was described by a monolayer adsorption capacity of 330 mg/g, which demonstrates the prospects of its use as a sorbent for water purification from various ions.

The relevance of water purification from manganese, iron, sulfate, and nitrite ions is confirmed by leading scientists [20–23]. Busarev et al. [20] showed that wastewater from many industrial enterprises is contaminated with heavy metal ions, including ferrous ions. Such wastewater has a highly acidic environment due to the use of mineral acids in the production. Iron ions in such conditions are toxic and can cause changes in biological organisms. Therefore, the purification of industrial wastewater from them is an important task. Lupandina et al. [21] showed that many types of industrial wastewater contain manganese ions. Such wastewater causes serious harm to aquatic and terrestrial flora and fauna. The hazard class of manganese ions is 3. The effect of manganese on the human body is manifested in persistent insomnia, decreased memory, and fatigue. Manganese affects the pulmonary and cardiovascular systems and causes allergic reactions. Poisoning of the human body and animals is classified by severity as poisoning with heavy metals. However, in this work, particulate sorbent-based resolutions are not posited within their investigation. Purolate's skill for the selective removal of manganese is experimentally presented via our work under batch and dynamic conditions.

Mikhailova and Popova [22] showed that wastewater from many industrial enterprises contains a significant concentration of sulfates formed when sulfuric acid is used in many industrial processes. Sulfate ions enter natural reservoirs and pose a serious threat to the environment. In humans, the ingestion of sulfates causes a laxative effect, dehydration, and irritable bowel syndrome. In addition, sulfates entering natural waters strongly salinate them, which harms aquatic plants and animals. Their investigation underscores ecological ramifications. But the study furnishes scant understanding since it neglects to delineate advanced absorbents able to eliminate sulfate. Our article investigation addresses such a deficiency by coupling X-ray diffraction (XRD) and X-ray fluorescence (XRF) for adsorption testing to illuminate the way anthracite-based carbon acts against sulfates.

Mikhailova and Garmashov [23] in their research found that an increase in the concentration of nitrates in groundwater is observed all over the world. Most often, nitrates enter reservoirs from the use of nitrogen fertilizers in agriculture. As agricultural land is depleted, farmers actively add nitrate fertilizers. Nitrates turning into nitrites, nitrosamines, and nitrosamides in drinking water can have carcinogenic potential. Their work stresses the health risks from nitrates along with nitrites. Nevertheless, their mechanistic elimination routes have not been investigated. Our investigation addresses this deficit via empirical assessment of purolate nitrite adsorption, also exploring intraparticle diffusion with surface chemistry for removal.

Studies confirm the urgency of removing Fe, Mn, SO₄²⁻, and NO₂⁻ from wastewater, but they generally stop at describing environmental impacts without linking contaminant toxicity to sorbent structure–performance relationships. Thus, the use of XRD analysis to study the physicochemical properties of sorbents based on the anthracite from the Kuzbass deposits (Russia) is performed for the first time. X-ray crystallography has primarily been used to determine the crystal structures of solids. However, relatively soft purolator crystals have a highly ordered crystal structure (hardness on the mineralogical scale 2.0–2.5) and function as a natural diffraction grating for X-rays.

The novelty of this investigation resides in the exhaustive depiction via diverse physicochemical techniques, which have infrequently been utilized in conjunction for Kuzbass-sourced anthracite-centered permeable carbon (purolate). Specifically, in this article, we used XRD via Rietveld refinement to resolve the graphite-dominated crystalline structure along with trace-phase composition. This differentiates this investigation from prior undertakings, which usually constrain themselves to gross adsorption skill. Sorbent action is represented more exhaustively via incorporating XRF elemental mapping, cryogenic N₂/Ar adsorption for textural scrutiny, and kinetic column trials under multi-ion circumstances. This methodological combination permitted us to connect structural attributes with the degree of efficient adsorption, thereby furthering our present comprehension of how anthracite-derived carbons purify water.

The purpose of this work was to study purolate (porous carbons from the Kuzbass deposits).

2 Methods and Materials

2.1 Objects of Research

The sorbent purolate (Sintez JSC, Rostov-on-Don, Russia) based on anthracites from Kuzbass deposits (Koks PJSC, Kemerovo region, Kuzbass, Russia) was the object of the research. Physicochemical characteristics of the sorbent are presented in Table 1.

Table 1. Characteristics of purolate

Characteristics	Units of Measurement	Values
Raw material	—	Anthracite
Graining (shape)	—	Crushed, irregular shape
Carbonation and activation	—	Single-step, steam-gas
Particle size	mm	0.1–3.0
Bulk density	g/cm ³	0.7–0.9
Endurance	%	90–95
Ash content	%	5
Moisture capacity	%	90
Specific surface area of pores (S _{BET})	m ² /g	311
Total pore volume by water	cm ³ /g	0.500
Pore volume		
Micro-	cm ³ /g	0.070
Meso-	cm ³ /g	0.000
Macro	cm ³ /g	0.430
pH of the water extract	—	8–9
Active Oxygen Content		
Acid type (COE OH [−]), general	mmol-eq/g	0.238
Phenolic –OH	mmol-eq/g	0.218
Carboxylic –COOH	mmol-eq/g	—
Lctone –COOH	mmol-eq/g	0.020
Basic type (COE H ⁺)	mmol-eq/g	0.120

Model aqueous solutions containing pollutants (iron, manganese, nitrates, sulfates, and their combinations) and monosolutions of these salts were the research objects as well. For the preparation of model solutions, salts containing m (Fe³⁺) = 0.7263 g, m (Mn²⁺) = 0.4324 g, m (SO₄^{2−}) = 44.3875 g, m (NO₂[−]) = 0.072 g dry salts separately (ferrous sulfate reagent grade, manganese sulphate reagent grade, sulfate ion according to the State Standard Reference Sample 7683-99 Interstate Standard Reference Sample 0198:2001 (1 g/dm³) and nitrite ion according to the State Standard Reference Sample 7753-2000 Interstate Standard Reference Sample 0202:2001 (1 g/dm³) (Ekroshim LLC, Moscow, Russia) were weighed on analytical scales (Mir LLC, Moscow, Russia) and placed in a beaker (Himbaza LLC, Moscow, Russia). 1 l of distilled water was measured with a 0.5 l measuring cylinder (Himbaza LLC, Moscow, Russia) and added to the salt. To better dissolve the salt, the solution was mixed with a 22 cm long glass rod with a diameter of 0.4 cm (Himbaza LLC, Moscow, Russia). The solution was poured into an adsorption container; the residues were drained into a prepared flask and stored in sealed dark glass flasks on steel shelves in a well-ventilated dry room at +20°C and a relative humidity of 60%–70%. The mass of the sorbent loaded into the column was 798 g.

For the preparation of monosolutions, each component was used separately and dissolved (m (Fe³⁺) = 0.7263 g, m (Mn²⁺) = 0.4324 g, m (SO₄^{2−}) = 44.3875 g, m (NO₂[−]) = 0.072 g) in 32 l of distilled water.

2.2 Thermogravimetric Analysis

Thermogravimetric analysis of the samples was performed with a synchronous thermal analysis device STA 449 C Jupiter by Netzsch (Melitek LLC, Moscow, Russia). Corundum crucibles were used to study the samples. The air supply rate to the sample chamber of the STA 449 C Jupiter thermal analyzer by Netzsch (Melitek LLC, Moscow, Russia) was 30 ml/min. The sample in the thermal analyzer chamber was heated to 1,000 °C at a rate of 10 °C/min. The selected range of 150, 1,000 °C reflects the adequate capture regarding thermal stability coupled with decomposition profiles, thus ensuring the onset of meaningful mass loss in carbonaceous sorbents and the upper limit at which anthracite undergoes major structural changes.

The experimental data were analyzed using the Netzsch Proteus Thermal Analysis software (Melitek LLC, Moscow, Russia), with the parameters recommended by the manufacturer of the thermal analyzer. To characterize the studied samples, the following indicators were used: *IT*—the ignition temperature, determined by the beginning of the curve bend on the thermogram during differential thermal analysis (DTA); *T*₁—the temperature of the beginning of mass loss, determined by the beginning of the curve bend of thermogravimetry (TG) and differential scanning

calorimetry (DSC); $T_{\max}$ —the temperature at which the maximum mass loss rate is reached, $V_{\max}$ —the maximum velocity at the inflection point, T_2 —the temperature of complete combustion of the sample. The coding of the studied sorbent samples is shown in Table 2.

Table 2. Encoding of the studied samples

Sample Name	Conditions of Receipt/Application
Purolate-Standard (base)	Semi-coke based on anthracite, which is manufactured using a single-step carbonation/air activation process
Purolate-Standard-1	Sorbent after adsorption of a mixture of ions (iron) from an aqueous solution
Purolate-Standard-2	Sorbent after adsorption of a mixture of ions (manganese) from an aqueous solution
Purolate-Standard-3	Sorbent after adsorption of a mixture of ions (sulfates) from an aqueous solution
Purolate-Standard-4	Sorbent after adsorption of a mixture of ions (iron, manganese, sulfates, nitrites) from an aqueous solution

2.3 X-Ray Diffraction Study of Purolate (Qualitative and Quantitative Analysis)

For obtaining diffractograms of the samples, the TD-3700 powder diffractometer (Tongda Science & Technology Co., Dandong, China) was utilized. The Bragg-Brentano method was used to focus a geometric vertical θ/θ goniometer of a linear semiconductor probe (Profpribor LLC, Voronezh, Russia) [24]. The radiation source was an X-ray tube (LLS LLC, St. Petersburg, Russia) with a copper anode and an average radiation wavelength $\text{CuK} = 0.15418 \text{ nm}$. Diffractograms were modeled using the Rietveld method on the TOPAS/XRD device (Bruker, Newtown Square, Pennsylvania, USA) to determine the phase composition [25]. Device parameters: rated power 32 kW, voltage range from 40 to 125 kV/150 kV, current range from 10 to 400 mA, angle range from 10 to 160 degrees.

2.4 X-Ray Fluorescence Examination of Purolate (Qualitative and Quantitative Analysis)

The sample was exposed to primary radiation from an X-ray tube (LLS JSC, St. Petersburg, Russia) to determine the intensity of the secondary fluorescent radiation emitted by the sample using a spectrophotometer Spectroscan Max-GVM (Spectron LLC, St. Petersburg, Russia). This was achieved by calculating the mass fraction of the element content based on a pre-established calibration characteristic [26]. The range of detected elements of the spectrometer is from Na to U. Detection limits (L) Na—0.1%, Mg—0.02%; from Al to P: 0.0005%–0.003%, from S to U: 0.0001%–0.0005%, The range of detectable contents from 3L to 100%; energy resolution 9 eV (Si $K\alpha$), 90 eV (Fe $K\alpha$); the voltage at the anode of the X-ray tube is 40 kV; the power of the X-ray tube is 160 W; does not require consumable gases and water (closed cooling cycle).

The degree of grinding was verified by passing the material through a specified sieve. 10 g was taken from the sorbent sample and dried in a drying cabinet or in the air. The material was ground to a size of 0.071 mm using a porcelain mortar. A sample of the material under investigation was placed in a cuvette. The sample covered polyethylene terephthalate film. It was secured with a clamping ring and casing and placed in a holder. Further, the holder with the sample was inserted into the spectrometer.

2.5 Obtaining Images of the Surface of the Purolate by Scanning Electron Microscopy in the 162 Mode of Detection of Secondary Electrons

To measure the linear dimensions of the purolator surface microrelief, including dielectric ones, a scanning electron microscope model JSM-6460LV manufactured by Microtrac LLC (St. Petersburg, Russia) was used in this study [27]. The microscope is composed of an electron-optical system, a secondary detector, a reflected electron detector, and object cameras with a mechanism for moving the objects. Additionally, there is a power supply unit, vacuum system, and video monitoring device [7]. When a focused electron beam was scanned over the surface of the purolite, images were acquired using a microscope. The brightness of the video monitor display was adjusted based on signals that were proportional to the number of secondary electrons detected [28]. The conditions of the analysis were as follows: the effective diameter of the electron probe in secondary electrons at 30 kV was not more than 30 nm, the measurement range of linear dimensions was 0.15, ..., 5,000 microns, the limits of the permissible relative error of linear dimensions were $\pm 5\%$, the range of acceleration voltage adjustment was 0.3, ..., 30 kV, the ambient temperature was $20 \pm 5^\circ\text{C}$, relative humidity was not more than 60%, atmospheric pressure was 84.07 kPa.

2.6 Extraction of Components with Purolate under Kinetic Conditions

An automated device, ASAP-2400 (Micromeritics Instrument Corporation, Norcross, Georgia, USA), which is controlled by a personal computer, measured the nitrogen adsorption at 77 K [29]. The device consisted of two distinct blocks. The first block was designed for the initial preparation of samples, which entailed heating the samples under controlled vacuum conditions generated using a pre-vacuum system. The heating process was conducted using

a furnace, with ampoules containing the samples placed inside. The temperature was maintained in the range from 20 to 350 °C, and the measurement accuracy was ± 1 °C. The pressure was measured using a vacuum gauge equipped with a thermocouple. Once a stable residual pressure below $\times 10^{-2}$ Torr was reached, it was determined that the process had completed.

After the preparation was complete, the ampoules were allowed to cool to room temperature. Further, they were filled with UHP nitrogen to atmospheric pressure and transferred to a second chamber to measure the adsorption isotherms. At the same time, independent measurements of isotherms were taken for two samples (the measurement unit is equipped with two ports). A capacitive pressure sensor was installed in a calibrated volumetric chamber and attached to each port. Isotherms were measured volumetrically by periodic dosing of nitrogen gas into the ampoule with the sample [29]. To correctly measure the adsorption isotherms, the volumes of the cold and hot zones of the ampoule were kept unchanged. To this end, for regulating the immersion level of ampoules in liquid nitrogen, the device was equipped with an automated control system. In order to reduce possible measurement errors associated with changes in ambient conditions (temperature and atmospheric pressure), the saturated vapor pressure corresponding to the temperature of liquid nitrogen was constantly measured. To accomplish this, each port was equipped with a suitable nitrogen thermometer (sealed tube), into which nitrogen gas was introduced at a predetermined interval of 120 min prior to condensation. After the system had reached equilibrium, the saturated vapor pressure was measured.

To study AC adsorption, 50 g of purolate was immersed in aqueous solutions of variable concentration. The total contact duration of the sorbent based on purolite with the solution was 24 h, and the continuous agitation time on the Wstazasarka Uniwersalna tup WU-4 apparatus (Bitlab LLC, Lublin, Poland) was 12 h. For the next 12 h, the system was under static conditions to establish a state of equilibrium between the solid adsorbent (purolate) and the absorbed substances (ions of solutions). Then the sample subsamples were filtered out for 6 h at room temperature through a “blue ribbon” filter with the following parameters: specific gravity 80 g/m², filtration time 180 s, approximate pore size 2–3 microns. The chosen pore size covers the mesoporous domain most relevant for ion adsorption from aqueous solutions, while also including the larger transition pores that influence intraparticle diffusion in dynamic systems.

We determined the equilibrium concentration of the adsorbent in the solution. The adsorption capacity of the purolate-based sorbent was measured by determining the difference between the initial and equilibrium concentration of the analyte in the solution. To calculate the equilibrium concentration of the adsorbent, we used Langmuir’s theory [30]. All experiments and calculations were carried out in three repetitions. The adsorption capacity of all sorbent types was determined by calculating the difference between initial and equilibrium solution concentrations, using the following formula [30]:

$$a = \frac{C_0 - C_{equ}}{m} V_{sol} \quad (1)$$

where, a —adsorption value (adsorption capacity, adsorbed amount [26]), g/g; C_0 —initial concentrations of pyridine in the solution, g/dm³; C_{equ} —equilibrium concentrations of pyridine in the solution, g/dm³; V_{sol} —the solution volume from which adsorption takes place, g/dm³; m —the sorbent subsample, g.

Considering that the main mechanism of adsorption for the studied ions in this study is ion exchange and complexation, Langmuir’s model of monomolecular adsorption [30] and Freundlich’s equation for an inhomogeneous surface [31] were used to analytically describe the adsorption isotherm and calculate adsorption parameters. The Freundlich model is applied to capture multilayer and heterogeneous surface adsorption typical for porous carbons. The Langmuir model tests the assumption of monolayer adsorption at energetically equivalent sites. These models allow relative estimation of uptake processes. Elucidation of equilibrium data is additionally stronger when utilizing both models. These models were applied in the region of medium adsorbate concentrations.

The Freundlich constant, k_F , has a mathematical meaning that is equal to the adsorption value at an equilibrium concentration of 1 (if c -g/dm³, then $k = a$) [31]. The specific characteristics of the adsorbent and the adsorbate define the adsorption constant. The constants in the Freundlich equation were derived by graphically solving a linear equation, which was obtained after taking the logarithm of the original equation:

$$\ln a = \ln k_F + \frac{1}{n} \ln C \quad (2)$$

According to Langmuir adsorption theory, adsorbate molecules adsorb at active sites that are always present on the surface of an adsorbent. These sites can be in the form of elevations, edges, or corners of crystal structures. By linearizing the equation derived from this theory, it is possible to determine the constants that appear in the equation:

$$\frac{1}{a} = \frac{1 + c}{aMaxK_c} \quad (3)$$

In most cases, a monolayer adsorption is not enough to fully compensate for the excess surface energy, and the effect of surface forces can extend to subsequent adsorption layers. Polymolecular adsorption can be represented as

the result of forced condensation of steam under the action of surface forces. In the Brunauer, Emmet, and Teller (BET) theory, the concept of the formation of “sequential adsorption complexes”, consisting of one, two, or more adsorbate molecules per adsorption site on the surface of the adsorbent, was added to the assumptions that form the basis for the Langmuir adsorption isotherm equation [32]. The equation of BET polymolecular adsorption looks as follows:

$$G = \Gamma \max \frac{p/p_s}{\left(1 - \frac{p}{p_s}\right) 1 + (K - 1)p/p_s} \quad (4)$$

where, K is the constant of the BET equation, p_s is the equilibrium pressure, Pa, p is the pressure at temperature T , K.

The main thermodynamic function, A , is the differential molar adsorption work maximum, which is equal to the negative of the Gibbs adsorption energy (ΔG):

$$A = -\Delta G = RT \ln K_c = RT \ln \left(\frac{C_s}{C} \right) \quad (5)$$

where, C_s —equilibrium concentration, g/dm³, C —concentration at temperature T , g/dm³, R —universal gas constant, equal to 8.314 J·mol⁻¹K⁻¹, T —temperature, K.

To assess the degree of wastewater treatment under dynamic conditions in a laboratory installation, we explored the possibility of extracting pollutants from solutions simulating the average composition of wastewater from coal mining operations [6, 8]. The model system consists of water and iron, manganese, sulfates, nitrite (50, 100, 10, 20 MAC (maximum acceptable concentration), accordingly). In a laboratory setup with a filter bed height of 0.15 m and a diameter of 0.05 m, the model solution was subjected to filtration. The column was packed with sorbent material, with a total pore volume of 0.5 cm³/g of sorbent. At room temperature, the sorbent was dried after pretreatment with distilled water until it reached an air-free state. The solution flow rate into the filter column was 1 dm³ per minute. All collected filtrates were analyzed for the presence of ions SO₄²⁻, NO₃⁻, Fe³⁺, and Mn²⁺.

Under the same experimental conditions, the efficiency of purification of individual substances in solutions was analyzed in dynamic conditions. The filtrate was analyzed for the residual ion content using the Iskroline 1,000 atomic absorption spectrometer (Iskroline Group of Companies, St. Petersburg, Russia). Spectrometer parameters: the presence of a DC arc, an AC arc, a DC pulse arc, an arc with variable polarity, a current range of 1–25 A, arc current stability, no more than 0.5%, a spectral range of 185–930 nm, detection limits of 10⁻⁵%–10⁻⁴%. The degree of purification of the solutions was determined by the formula:

$$C = \frac{C_{ini} - C_{fin}}{C_{ini}} 100\% \quad (6)$$

where, C_{ini} , C_{fin} are the initial and final concentrations of the dispersed phase in an aqueous medium, g/dm³, respectively [33].

2.7 Extraction of Components with Purolate under Dynamic Conditions

The installation consisted of a system for the preparation of a solution to simulate water pollution [34] (including two water tanks), a pump for supplying the solution to the treatment system, a network of pipes and shut-off valves, a multi-stage filter, and a multi-channel automatic data collection board; the measurement and software control of the experiment was connected to a computer. The first stage of the system is a preliminary filtration unit with a pore size of 5 microns. The purpose of this stage was to remove any relatively large undissolved particles that may be present in the water and could potentially clog the subsequent stages of the process. The second stage of the filtration process involves the use of a sorption material, which serves as the secondary filter. The installation is shown in Figure 1.

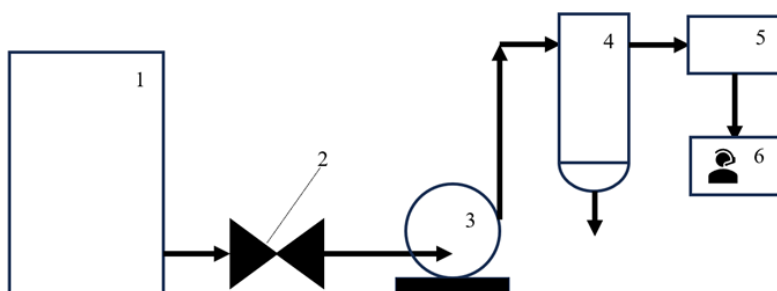


Figure 1. Laboratory installation for purifying solutions using purolate: 1—water tanks; 2—shut-off valves; 3—pump; 4—universal step filter; 5—control panel; 6—data collection panel connected to a computer

3 Results and Discussion

3.1 X-Ray Diffraction Analysis of the Purolate Sorbent

The results of XRD analysis of the purolate sorbent are presented in Table 3.

Table 3. X-ray diffraction (XRD) analysis of the Purolate-Standard (base)

Sorbent Composition	Composition
MgO (%)	0.2
Al ₂ O ₃ (%)	0
SiO ₂ (%)	6
P ₂ O ₅ (%)	0.08
K ₂ O (%)	0.16
CaO (%)	0.54
TiO ₂ (%)	0.09
Fe ₂ O ₃ (%)	2.31
MnO (mg/kg)	119
V (mg/kg)	3
Cr (mg/kg)	8
Co (mg/kg)	51
Ni (mg/kg)	48
Cu (mg/kg)	541
Zn (mg/kg)	5,346.8
As (mg/kg)	20
Sr (mg/kg)	304
Pb (mg/kg)	41
Rb (mg/kg)	0
Zr (mg/kg)	0
Ba (mg/kg)	256

It follows from the data in Table 4 that, in addition to carbon and oxygen, significant amounts of Zn (5,346.8 mg/kg), Ba (256 mg/kg), Sr (304 mg/kg), Cu (541 mg/kg), and MnO (119 mg/kg) were detected in the purolate. However, it did not contain Al₂O₃, SiO₂, Rb, and Zr.

Table 4. X-ray diffraction (XRD) analysis of the organic sorbents

Sorbent Composition	Sorbents							References
	I	II	III	IV	V	VI	VII	
MgO (%)	0.2	0.2	0.1	0.1	0.1	0.1	6.2	[35, 36]
Al ₂ O ₃ (%)	0.0	1.0	0.5	0.0	1.0	1.0	0.0	
SiO ₂ (%)	6.0	46.8	8.6	5.9	52.7	46.8	16.9	
P ₂ O ₅ (%)	0.08	0.073	0.234	0.107	0.063	0.073	0.064	
K ₂ O (%)	0.16	0.72	0.12	0.48	0.64	0.72	0.0	
CaO (%)	0.54	0.60	2.16	0.59	0.60	0.60	0.34	
TiO ₂ (%)	0.09	0.52	0.29	0.21	0.42	0.52	0.02	
Fe ₂ O ₃ (%)	2.31	6.12	3.54	1.59	3.84	6.12	9.53	
MnO (mg/kg)	119	350	530	78	261	350	1,749	
V (mg/kg)	3	173	0	170	105	173	0	[35, 37]
Cr (mg/kg)	8	177	26	55	131	177	1352	
Co (mg/kg)	100	33	67	54	22	33	51	
Ni (mg/kg)	48	64	49	58	48	64	1,136	
Cu (mg/kg)	541	68	66	71	49	68	863	[35, 38]
Zn (mg/kg)	5,346.8	117.0	106.0	71.0	62.0	117.0	69.0	
As (mg/kg)	20	14	5	2	14	14	0	
Sr (mg/kg)	851	71	304	122	72	71	24	
Pb (mg/kg)	45	20	40	41	15	20	27	[39]
Rb (mg/kg)	0	69	0	0	61	69	4	
Zr (mg/kg)	0	67	20	0	71	67	0	
Ba (mg/kg)	849	285	256	457	301	285	0	

Note: I—Purolate; II—AG-3; III—SKD-515; IV—KAU; V—ODN-2F; VI—AS; VII—MS.

These contaminants are not inactive: Zn can alter surface charge and foster electron-transfer reactions, while Fe^{3+} and Mn^{2+} might encounter rivalry from Ba and Sr for adsorption sites. They exist to aid in explaining subsequent observations concerning differing selectivity among single-ion and multi-ion experiments. Minute constitution exerts a key function within the veritable operation concerning anthracite-based absorbent.

In the study of the published data, the physicochemical properties of other natural sorbents based on AC were established. The results of the analysis of the published and received data are summarized in Tables 4 and 5.

Table 5. Characteristics of organic sorbents

Characteristics	Units of Measurement	Sorbents						
		I [36]	II [39]	III [39]	IV [38]	V [37]	VI [40]	VII [37]
Raw material	–	Anthracite	Coal grade D, wood chemical and coke chemical resin	A mixture of hard coals of grades K and D, coke chemical resin	Coconut shells	Opoka	Aluminosilicate	Aluminosilicate
Graining (shape)	–	Crushed, irregular shape	Granular, cylinder	Granular, cylinder	Crushed, irregular	–	–	–
Carbonation and activation	–	Single-step, steam-gas	Two-stage, vapour-gas	Two-stage, vapour-gas	Two-stage, vapour-gas	–	–	–
Particle size	mm	0.1–3.0	1–1.5	0.5–1.5	0.4–1.7	0.3–1.5	0.7–1.4	0.7–1.4
Bulk density	g/cm^3	0.7–0.9	0.4–0.55	0.52	0.49	0.68–0.72	0.46–0.49	1.35–1.94
Endurance	%	90–95	75–85	75	98	99.9	99.94	99.9
Ash content	%	5	1–1.5	–	<6	–	–	–
Moisture capacity	%	90	92	95	92	80–95	–	–
Specific surface area of pores (S_{BET})	m^2/g	311	810	791	1512	120–180	–	–
Total pore volume by water	cm^3/g	0.500	0.880	0.620	0.951	–	–	–
Pore Volume								
Micro-	cm^3/g	0.070	0.271	0.360	0.622	–	0.303	–
Meso-	cm^3/g	0.000	0.059	0.200	0.110	0.633	0.067	0.063
Macro	cm^3/g	0.430	0.531	0.061	0.219	0.004	–	0.516
pH of the water extract	–	8–9	6.5	7.6	7.8	6–7	8.3	7–8
Active Oxygen Content								
Acid type, general (COE OH^-)	mmol-eq/g	0.238	0.395	0.338	0.344	–	–	–
Phenolic –OH	mmol-eq/g	0.218	0.321	0.181	0.194	–	–	–
Carboxylic –COOH	mmol-eq/g	–	0.035	–	0.090	–	–	–
Lactone –COOH	mmol-eq/g	0.020	0.039	0.157	0.060	–	–	–
Basic type (COE H^+)	mmol-eq/g	0.120	1.434	0.571	0.778	–	–	–

Note: I—Purolate; II—AG-3; III—SKD-515; IV—KAU; V—ODN-2F; VI—AS; VII—MS.

When comparing the physicochemical properties of AC sorbents presented in Table 6, we found that purolate had

the largest particle size range (from 0.1 to 3 mm). Other organic sorbents, according to published data, had pores ranging from 0.5 to 1.7 mm. However, the total water pore volume of purolate was the smallest -0.5 cm³/g. It was shown that, unlike the structure of other AC sorbents, there were no mesopores in the structure of purolate. The data (Table 6) indicate that purolate has the largest pH range of aqueous extract, 8–9 units.

Table 6. Phase composition of purolate standard (base) according to the modeling results

Graphite 2H (icsd_53781)	SiO ₂ -Quartz (icsd_174)	Fe _{3.7} H ₅ O ₉ Si (icsd_161270)	(FeO) _{0.198} (MnO) _{0.802} (icsd_60689)	Ca(Fe ²⁺ , Mg)(CO ₃) ₂ (icsd_100417)
~95%	<2%	<2%	<2%	<1%

The data in Table 7 shows that purolate contained the largest amounts of Zn (5,346.8 mg/kg), As (20 mg/kg), Sr (851 mg/kg), Pb (45 mg/kg), Ba (849 mg/kg), and Co (100 mg) among all analyzed sorbent samples/kg). However, compounds such as Al₂O₃, Rb, and Zr were absent in purolate.

Table 7. Thermogravimetric analysis of purolate samples

Samples	T ₁ max (°C)	V ₁ max (%/min)	V ₂ max (°C)	V ₂ max (%/min)	Δm (%)			
					A	B	C	D
0	410	0.10	651	0.23	0.7	5.6	0.2	5.9
1	427	0.13	662	0.28	1.0	5.9	0.4	6.3
2	431	0.16	654	0.33	0.8	7.2	1.5	8.7
3	418	0.14	664	0.35	0.8	0.9	0.5	7.2
4	—	—	—	—	8.7	1.0	0.1	9.8

Note: Samples 0—Purolate-Standard (base); 1—Purolate-Standard-1; 2—Purolate-Standard-2; 3—Purolate-Standard-3; 4—Purolate-Standard-4.

A—20–150 °C; B—150–700 °C; C—700–1000 °C; D—150–1,000 °C.

3.2 Results of Electron Microscopy

As a result of the study of purolate by scanning electron microscopy, a microimage of the initial sorbent was obtained (Figure 2).

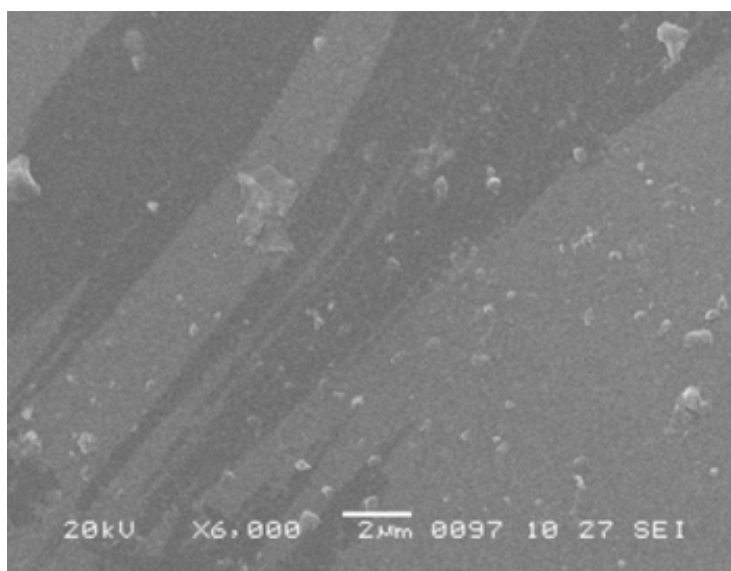


Figure 2. Results of electron microscopy of purolate before adsorption (magnification 6,000×)

3.3 X-Ray Diffraction Results

Using the Rietveld technique in the Topas software (Bruker), the diffraction patterns were simulated to determine the phase composition (Figures 3 and 4). The initial structural information was obtained from the Inorganic Crystal Structure Database (ICSD). According to the results of the study of model solutions, the phase composition of purolate was within the errors of both the XRD method (usually at least 5% of the substance is required) and the statistical sampling (Table 6).

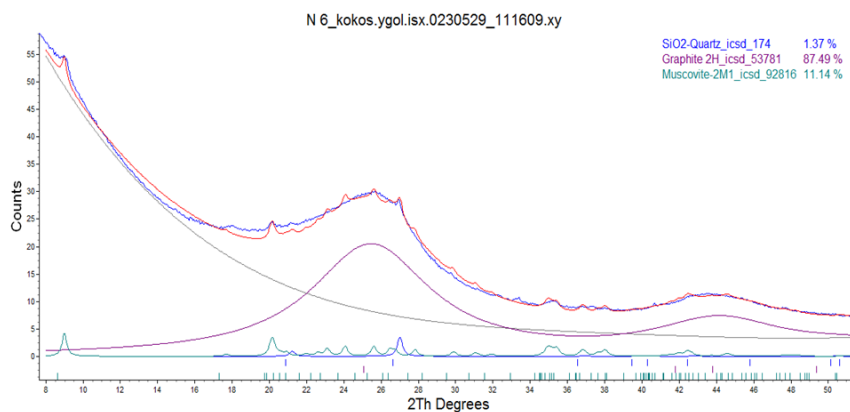


Figure 3. Modeling of the diffraction pattern by the Rietveld method in the Topas program. Purolate (initial sorbent)

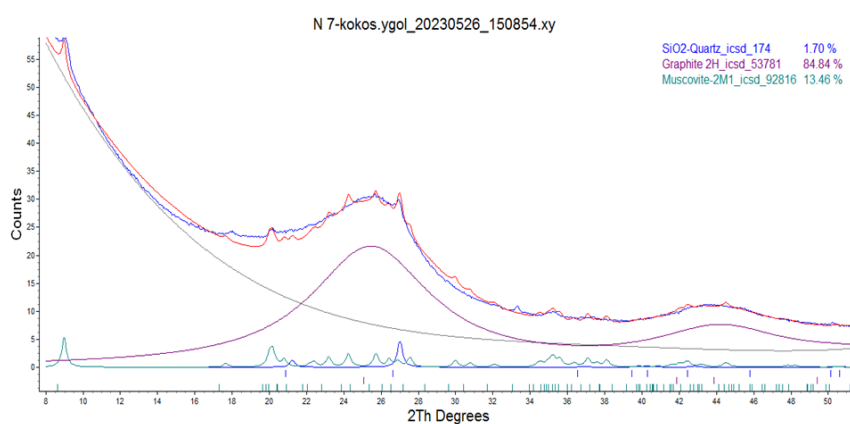


Figure 4. Modeling of the diffraction pattern by the Rietveld method in the Topas program. Purolate (sorbent after adsorption of a mixture of ions (iron, manganese, sulfates, nitrites) from a solution)

The data (Table 6) indicate that, according to the results of the study of model solutions, purolate contains Graphite 2H (icsd_53781) phase the most, 95%. Before use, the sorbent samples were thermally activated by calcination at 1,000 °C with chloride and sodium carbonate. Calcined minerals acquired the property of not swelling in water.

3.4 Thermogravimetric Analysis

The results of thermogravimetric analysis of purolate samples conducted in an oxidizing atmosphere are presented in Table 7.

The mass loss associated with the release of hygroscopic moisture was observed when all samples were heated to a temperature of about 200 °C, which may be due to their larger specific surface area. The main decomposition of organic matter took place in the temperature range from 500 to 1,000 °C (Table 7). The maximum decomposition temperature ($T_{\max}$) of the samples varied as a result of their use in sorption processes. The Purolate-Standard-2 sample, which was used to adsorb iron ions from an aqueous solution, is assumed to be the most promising for research, since it was activated at the highest temperature $T_{\max}$.

Analysis of the data obtained using thermogravimetric analysis showed that purolate samples Purolate-Standard-1, Purolate-Standard-2, and Purolate-Standard-3 had the greatest thermal stability. Up to a temperature of 150 °C, the thermograms of these three samples showed mass loss associated with the release of hygroscopic moisture, the largest value of which was determined for sample 2 (8.7%) (Table 7). In the temperature range of 150–700 °C, the main mass loss of the samples occurred. However, its value for the Purolate-Standard-3 and Purolate-Standard-4 samples was insignificant, about 1%. The Purolate-Standard-2 sample was characterized by the highest value (7.2%). The mass loss of this sample occurred in two stages. Two peaks of mass loss at different rates ($T_{1\max}$ and $T_{2\max}$) were observed on the thermogravimetric analysis curve (Table 7 and Figure 5). After using the Purolate-Standard-1 sorbent in adsorption processes (Purolate-Standard-3 and Purolate-Standard-4 samples), there was a tendency for the first maximum temperature ($T_{1\max}$) and the second maximum temperature ($T_{2\max}$) to decrease on the thermogram curve.

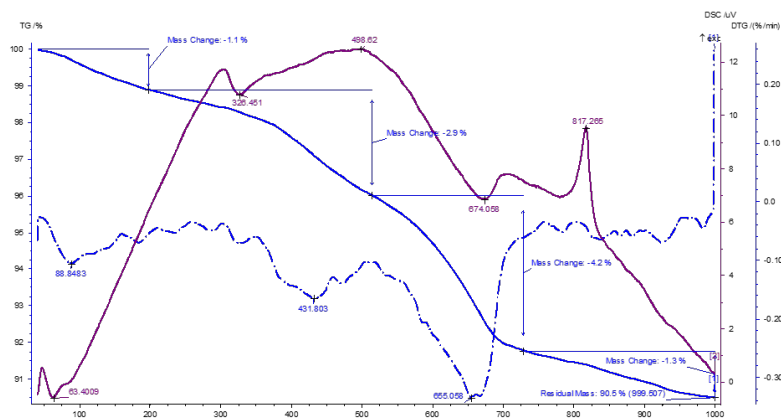


Figure 5. Thermogram of the Purolate-Standard-1 sample

Figure 6 shows the adsorption isotherms N_2 (77 K) for the Purolate-Standard-1, Purolate-Standard-2, Purolate-Standard-3, and Purolate-Standard-4 samples.

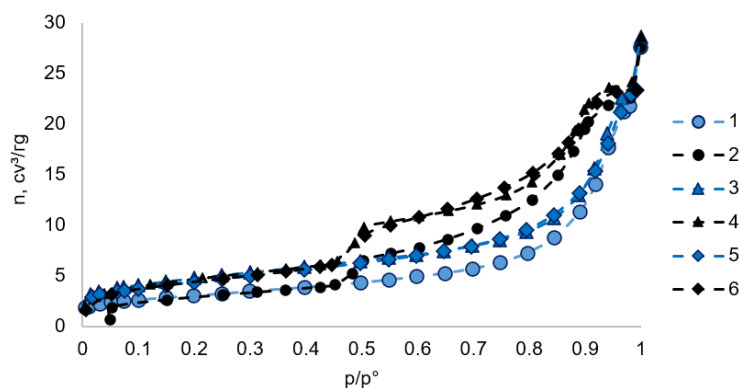


Figure 6. Adsorption isotherms N_2 (77 K) of the Purolate-Standard-1, Purolate-Standard-3, and Purolate-Standard-4 samples: 1—adsorption, Purolate-Standard-1; 2—desorption, Purolate-Standard-1; 3—adsorption, Purolate-Standard-3; 4—desorption, Purolate-Standard-3; 5—adsorption, Purolate-Standard-4; 6—desorption, Purolate-Standard-4

Figure 7 shows the pore size distribution curves in the Purolate-Standard-1, Purolate-Standard-3, and Purolate-Standard-4 samples.

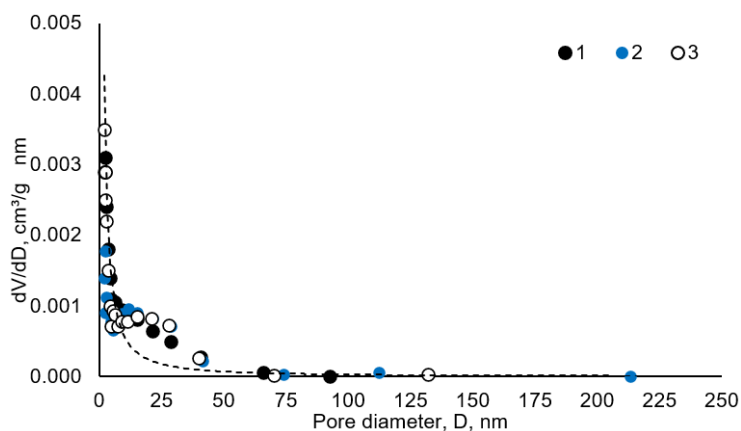


Figure 7. Pore size distributions: 1—Purolate-Standard-1; 2—Purolate-Standard-3; 3—Purolate-Standard-4 samples

Figure 8 shows the adsorption isotherms N_2 (77 K) of the Purolate-Standard-2 sample.

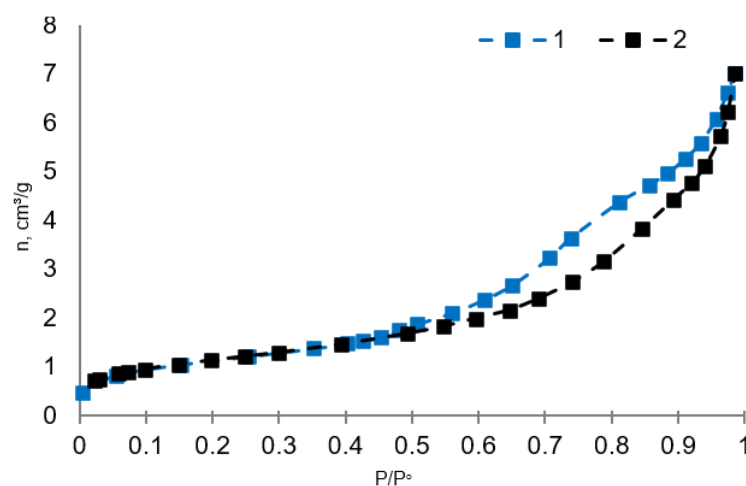


Figure 8. Adsorption isotherms N₂ (77 K) of the Purolate-Standard-2 sample: 1—adsorption; 2—desorption

Figure 9 shows the adsorption isotherms N₂ (77 K) of the Purolate-Standard-2 sample.

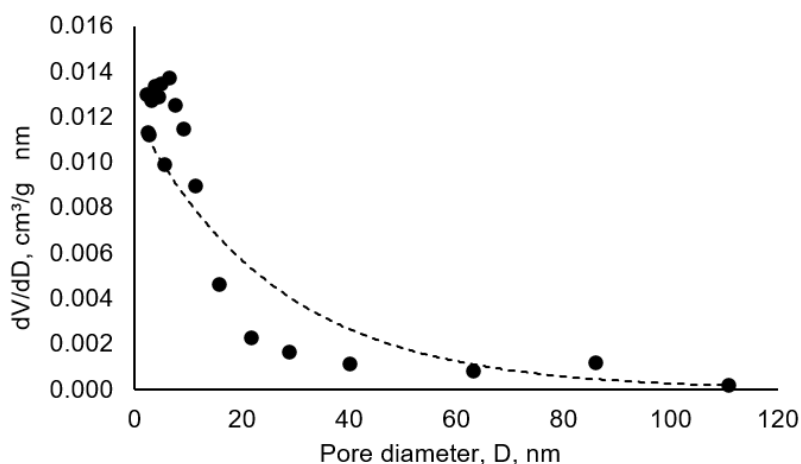


Figure 9. Pore size distribution in the Purolate-Standard-2 sample

The textural characteristics of the samples are presented in Table 8.

Table 8. Values of the textural characteristics of the purolate samples according to the adsorption data of N₂ and Ar at 77 K

Samples	a_S (m ² /g)	C_{BET}	a_{meso} (m ² /g)	V_{micro} (cm ³ /g)	V_{sum} (cm ³ /g)
0	9.4 ± 2.8	101 ± 30.3	7.5 ± 2.3	0.001 ± 0.0003	0.021 ± 0.006
1	11.0 ± 3.3	142 ± 42.6	8.4 ± 2.5	0.001 ± 0.0003	0.029 ± 0.009
2	16.8 ± 5.0	161 ± 48.3	12.6 ± 3.8	0.002 ± 0.0006	0.032 ± 0.009
3	15.2 ± 4.6	195 ± 58.5	14.0 ± 4.2	0.001 ± 0.0003	0.030 ± 0.009
4	93.0 ± 27.9	86 ± 25.8	85.0 ± 25.5	0.003 ± 0.0009	0.185 ± 0.056

Note: Samples 0—Purolate-Standard (base); 1—Purolate-Standard-1; 2—Purolate-Standard-2; 3—Purolate-Standard-3; 4—Purolate-Standard-4.

The data from Table 8 shows that the highest values of a_S and C_{BET} corresponded to the Purolate-Standard-2 sample, the highest values of a_{meso} (14.0 m²/g) corresponded to the Purolate-Standard-3 sample, and the highest values of V_{micro} (0.003 cm³/g) and V_{sum} (0.185 cm³/g) corresponded to the Purolate-Standard-4 sample.

3.5 Extraction of Manganese and Iron Ions under Kinetic Conditions from Monosolutions

During the extraction of contaminants from monosolutions under kinetic conditions, the data presented in graphical form were obtained [40]. When extracting manganese from an aqueous solution at a concentration of 0.0004 g/dm³, the dependence shown in Figure 10 was obtained.

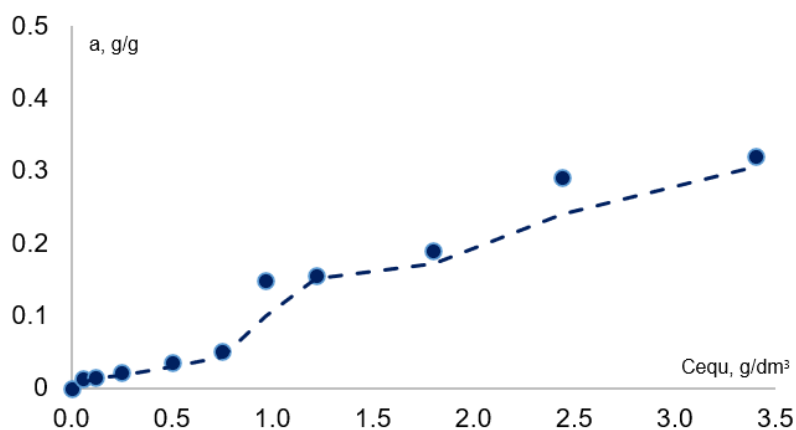


Figure 10. Isotherm of manganese (II) adsorption from aqueous solutions with purolate

The parameters f manganese adsorption were determined using the Freundlich and Langmuir models, which describe the monomolecular adsorption of the adsorbate onto active sites. The adsorption isotherms, in the linearized coordinates, are presented in Figure 11.

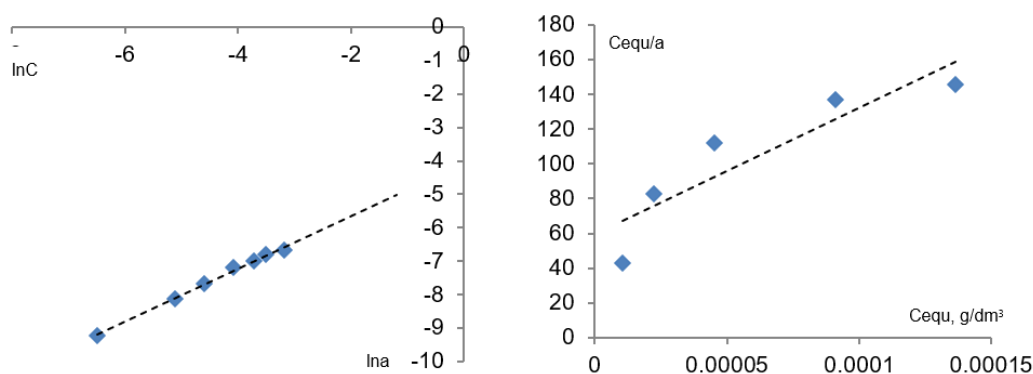


Figure 11. Isotherms of manganese adsorption from aqueous solutions in coordinates: (a) Freundlich; (b) Langmuir

For the models, we calculated the theoretical adsorption isotherms for manganese. A comparative analysis of the experimental and theoretical adsorption isotherms (Figure 12) showed that for all the studied systems, the Langmuir and Freundlich models could adequately describe their isotherms.

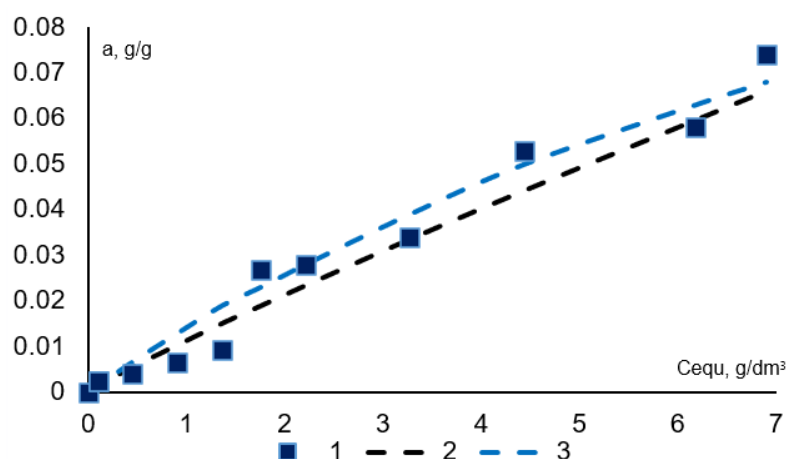


Figure 12. Experimental isotherms of manganese adsorption from aqueous solutions with purolate: 1—experimental data; 2—calculated data, Freundlich equation; 3—calculated data, Langmuir equation

The isotherm of the adsorption of iron ions from aqueous solutions on the carbon sorbent purolate is shown in Figure 13.

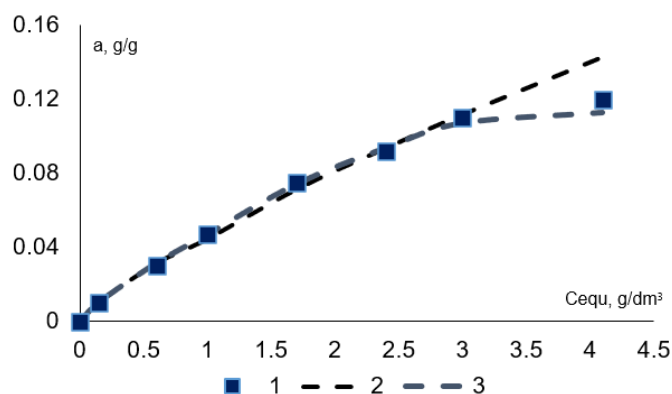


Figure 13. Isotherm of adsorption of iron ions from aqueous solutions with purolate: 1—experimental data; 2—calculated data, Freundlich equation; 3—calculated data, Langmuir equation

A comparative analysis of the experimental and theoretical adsorption isotherms (Figures 12, 13) demonstrated that for all the studied systems, the Langmuir and Freundlich models could adequately describe their isotherms. The adsorption parameters of manganese and iron are presented in Table 9.

Table 9. Adsorption characteristics of sample 1 (purolate standard) during the adsorption of monosolutions

Sorbent Type	Freundlich Equation			Langmuir Model		
	$1/n$	k (g/g)	R^2	$a_{\max}$ (g/g)	$-G$ (kJ/mol)	R^2
Iron	0.79 ± 0.24	0.018 ± 0.005	86.5 ± 25.95	0.14 ± 0.04	17.5 ± 5.25	96.01 ± 28.80
Manganese	0.90 ± 0.27	0.005 ± 0.001	93.8 ± 28.14	1.05 ± 0.32	26.09 ± 7.83	90.36 ± 27.11

The data (Table 10) show that the $1/n$ constant of the Freundlich equation for the extraction of iron ions had a smaller value (0.79) than that of manganese ions (0.90). However, the k constant for iron ions was significantly higher than the k constant for manganese ions (0.018 and 0.0048, respectively). The adsorption value according to the Langmuir equation was higher for manganese ions (1.05 mg/g). The Gibbs energy had a similar dependence: for manganese ions, it is greater than for iron ions.

Table 10. Sample analysis results for the content of sulfate ions after adsorption of the model mixture and monosolutions on Purolate-Standard-1

Purification Cycles	Model Mix		Monosolution of Sulfate Ions	
	D_{avg}	C (mg/l)	D_{avg}	C (mg/l)
1.0	0.0873 ± 0.03	568.4 ± 170.52	0.0523 ± 0.02	789.6 ± 236.88
1.1	0.0436 ± 0.01	426.8 ± 128.04	0.0464 ± 0.01	601.4 ± 180.42
1.2	0.0448 ± 0.01	436.2 ± 130.86	0.0504 ± 0.02	735.2 ± 220.56
2.0	0.0285 ± 0.01	389.6 ± 116.88	0.0457 ± 0.01	706.2 ± 211.86
2.1	0.0336 ± 0.01	348.2 ± 104.46	0.0518 ± 0.02	654.2 ± 196.26
2.2	0.0389 ± 0.01	309.6 ± 92.88	0.0418 ± 0.01	621.3 ± 186.39
3.0	0.0346 ± 0.01	423.8 ± 127.14	0.0502 ± 0.02	782.8 ± 234.84
3.1	0.0431 ± 0.01	355.8 ± 106.74	0.0495 ± 0.01	777.4 ± 233.22
3.2	0.0321 ± 0.01	336.5 ± 100.95	0.0361 ± 0.01	661.8 ± 198.54
4.0	0.0383 ± 0.01	400.8 ± 120.24	0.0596 ± 0.01	737.8 ± 221.34
4.1	0.0403 ± 0.01	388.8 ± 116.64	0.0475 ± 0.01	654.4 ± 196.32
4.2	0.0291 ± 0.01	312.8 ± 93.84	0.0298 ± 0.01	610.3 ± 183.09
5.0	0.0336 ± 0.01	466.6 ± 139.98	0.0465 ± 0.01	702.1 ± 210.63
5.1	0.0441 ± 0.01	430.6 ± 129.00	0.0444 ± 0.01	598.8 ± 179.64
5.2	0.0487 ± 0.01	377.8 ± 113.34	0.0536 ± 0.01	580.4 ± 174.12

3.6 Extraction of Nitrite Ions under Kinetic Conditions from Monosolutions

The results of the extraction of nitrite ions from a monosolution at a concentration of SV 400 MAC using samples of the purolate sorbent are presented in Table 11 and Figure 14.

Table 11. Sample analysis results for the content of nitrite ions after adsorption of the model mixture and monosolution on Purolate-Standard-1

Purification Cycles	Model Mix	Monosolution of Nitrite Ions
	C (mg/l)	C (mg/l)
1.0*	2.069 ± 0.62	2.431 ± 0.73
1.1	1.605 ± 0.48	2.246 ± 0.67
1.2	1.874 ± 0.56	1.649 ± 0.49
2.0	1.192 ± 0.36	1.829 ± 0.55
2.1	0.996 ± 0.30	1.743 ± 0.52
2.2	1.102 ± 0.33	1.798 ± 0.54
3.0	1.148 ± 0.34	1.906 ± 0.57
3.1	1.141 ± 0.34	1.909 ± 0.57
3.2	1.027 ± 0.31	1.912 ± 0.57
4.0	1.148 ± 0.34	1.835 ± 0.55
4.1	0.996 ± 0.30	1.804 ± 0.54
4.2	0.989 ± 0.30	1.772 ± 0.53
5.0	0.952 ± 0.30	1.794 ± 0.54
5.1	0.952 ± 0.30	1.745 ± 0.52
5.2	0.828 ± 0.25	1.745 ± 0.52

Note: *There were three stages in each of the five purification cycles.

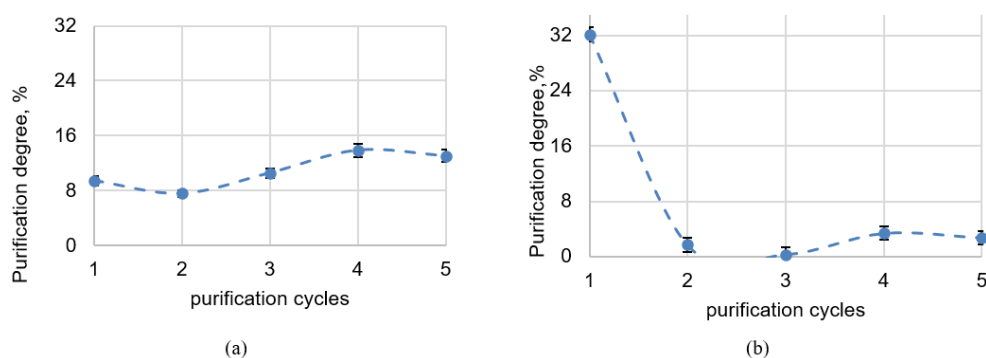


Figure 14. Output curves of the degree of removal of nitrite ions from (a) the model mixture and (b) the monosolution

The degree of purification from nitrite ions on purolate by the end of cycle 5 was 60.0% of the model mixture and 28.2% of the monosolution.

3.7 Analysis of the Degree of Purification from Sulfate Ions under Dynamic Conditions

The results of the extraction of sulfate ions after adsorption of the model mixture and monosolutions using the purolate sorbent are presented in Table 10 and Figure 15.

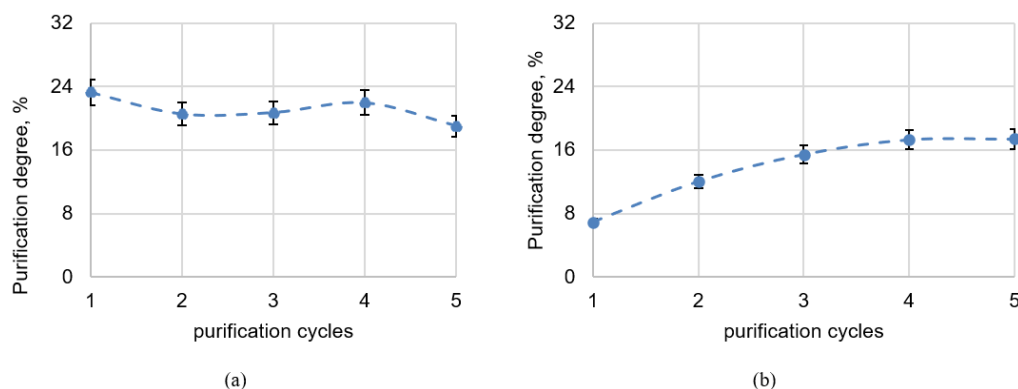


Figure 15. Output curves of the degree of purification from sulfate ions by purolate sorbent in (a) a model mixture of wastewater and (b) an individual solution

The analysis of graphical dependencies (Figure 15) showed that the degree of purification from 1–5 cycles for the extraction of sulfate ions from the model mixture practically did not change and amounted to 20%–24%. In the case of extraction from a monosolution, the degree of extraction of sulfate ions increased with each cycle and reached 18%. In total, for 5 cycles, the degree of purification was 33.5% for sulfate ions from the model mixture and 26.5% from the monosolution.

The isotherm of adsorption [40] of iron, nitrite ions, and sulfate ions from aqueous solutions with purolate is shown in Figures 16–18, respectively.

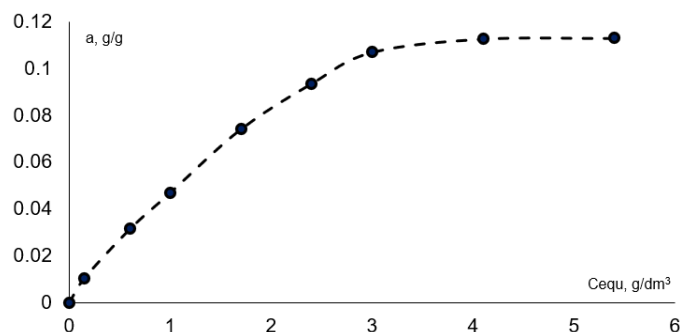


Figure 16. Isotherm of iron ions adsorption from aqueous solutions with purolate

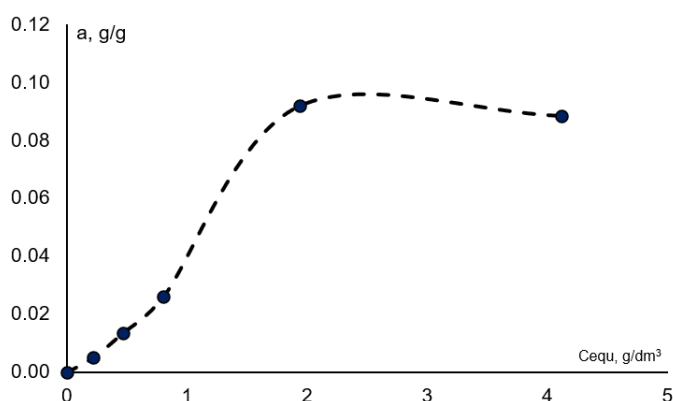


Figure 17. Isotherm of nitrite ions adsorption from aqueous solutions with purolate

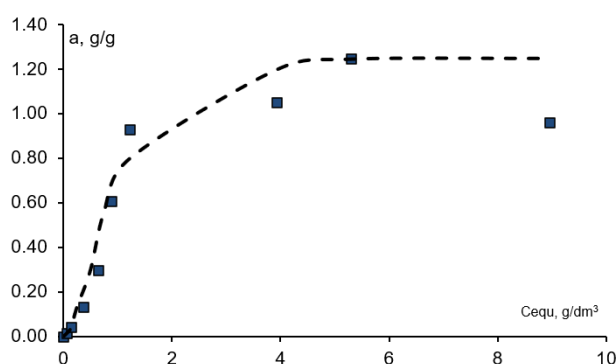


Figure 18. Isotherm of sulfate ions adsorption from aqueous solutions with purolate

The analysis of Figures 10, Figures 16–18 indicates that when using purolate to isolate manganese ions from aqueous solutions, the highest adsorption value (0.3 g/g) was observed for C_{equ} of 3.5 g/dm³. The highest adsorption was at the extraction of iron ions 0.14 g/g with C_{equ} of 4.0 g/dm³, at the extraction of nitrite ions 0.1 g/g with C_{equ} of 2.0 g/dm³, and at the extraction of sulfate ions 1.2 g/g with C_{equ} of 6.0 g/dm³.

The kinetic curve of ion extraction and the dynamics of the degree of achievement of adsorption equilibrium during the extraction of manganese, iron, nitrite ions, and sulfate ions are shown in Figures 19–22.

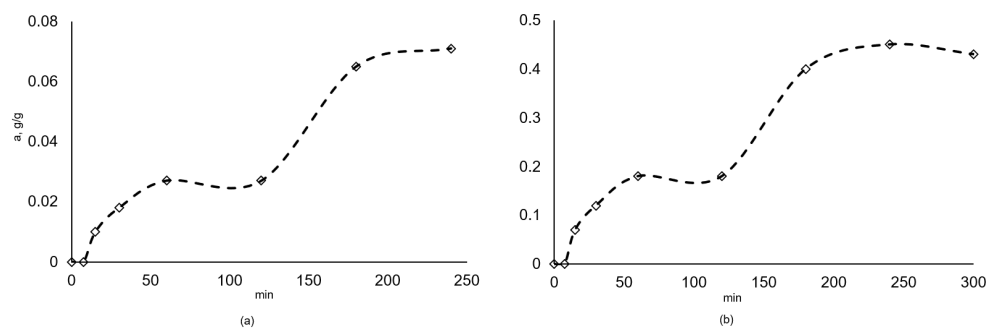


Figure 19. The kinetic curve of ion extraction (a) and the dynamics of the degree of achievement of adsorption equilibrium γ (b) during the release of manganese

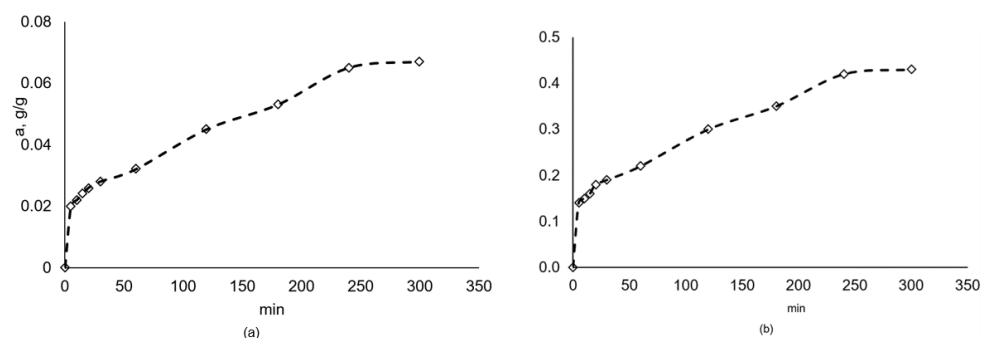


Figure 20. The kinetic curve of ion extraction (a) and the dynamics of the degree of achievement of adsorption equilibrium γ (b) during the release of iron ions

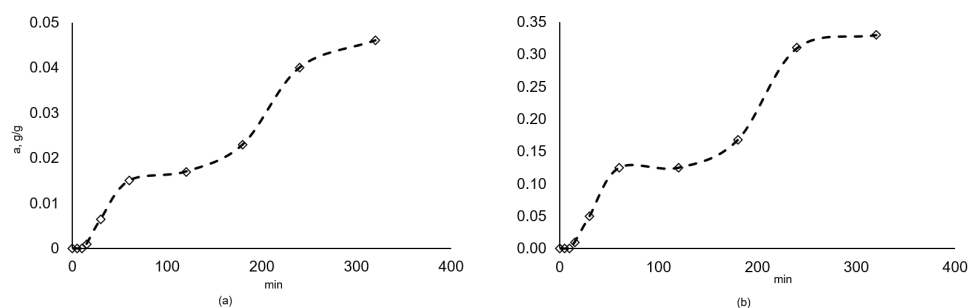


Figure 21. The kinetic curve of ion extraction (a) and the dynamics of the degree of achievement of adsorption equilibrium γ (b) during the release of nitrite ions

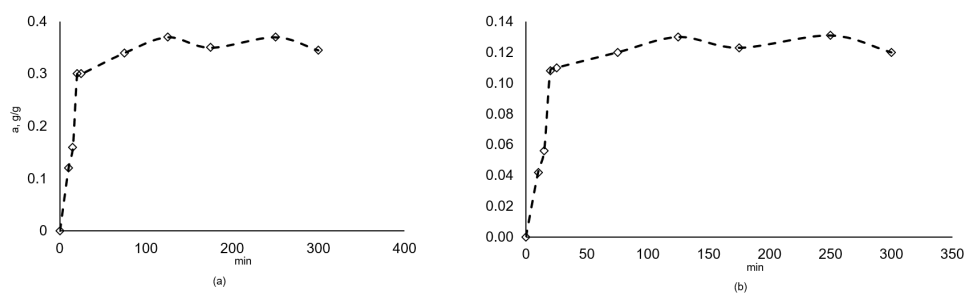


Figure 22. The kinetic curve of ion extraction (a) and the dynamics of the degree of achievement of adsorption equilibrium γ (b) during the release of sulfate ions

Based on the analysis of the kinetic curves of ion extraction (Figures 19–22), when extracting manganese ions, the greatest adsorption (0.07 g/g) was observed at 250 min; when extracting iron ions, the greatest adsorption (0.065 g/g) was observed at 300 min; when extracting nitrite ions, the greatest adsorption (0.045 g/g) was observed at 300 min; when extracting sulfate ions, the highest adsorption was 0.035 g/g with an extraction time of 300 min.

The breakthrough curve of the system provides information on the service life of the layer and the regeneration time required for the sorbent. The shape of the breakthrough curve is dependent on the inlet flow rate, concentration, column diameter, bed height, and physicochemical characteristics of the sorbent material. A fixed layer of a given size can absorb dissolved substances entering the layer, which is equivalent to the adsorption of pollutants not only until equilibrium is reached, but also depending on the rate of adsorption [35]. The adsorption breakthrough curve is shown in Figure 23. The data of the figure indicate that the service life of the sorbent layer was 380 min at an adsorption temperature of 28–30 °C.

The resulting absorption isotherm was processed by analyzing various parts of the isotherm, as well as calorimetry over the entire pressure range [38].

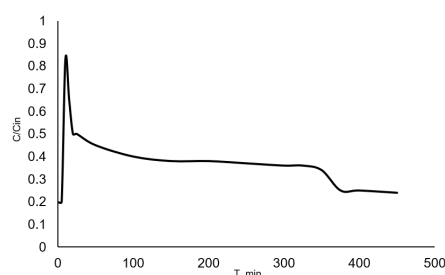


Figure 23. Adsorption breakthrough curve

We determined the presence, size, and number of pores by analyzing the photographs of the surface of the sorbent material obtained using a scanning electron microscope (Figure 2).

All sorbent samples before and after adsorption (Table 3) are almost identical. After adsorption, the phase composition and structure of the purolate did not change significantly. Along with graphite-like carbon, the crystalline phase SiO_2 -Quartz was observed everywhere, and weak reflexes of the remaining phases may indicate the presence of phases based on the structures $\text{Fe}_{3.7}\text{H}_5\text{O}_9\text{Si}$ (icsd_161270), $\text{Ca}(\text{Fe}^{2+}, \text{Mg})(\text{CO}_3)_2$ (icsd_100417), and $(\text{FeO})_{0.198}(\text{MnO})_{0.802}$ (icsd_60689). The mass fraction of each impurity did not exceed 2.0%, which is on the verge of the sensitivity of the XRD method, and an accurate determination of the content is not possible. Other phases may be present, as not all reflexes were identified. The chemical composition (stoichiometry) of these phases may differ from those given, since the structures could be defective and contain impurity elements [41].

From an environmental perspective, purolate's ability for maintenance regarding removal efficiency under contaminant load variations is suggested via the observed pseudo-second-order kinetics, as well as mixed chemisorption intraparticle diffusion mechanisms. This suggests practical invulnerability during treatment for mine-impacted effluents. It also suggests resilience regarding the treatment of industrial effluents when ion concentrations vary. Comparatively rapid primary absorption advantages motile sieving, while tardier permeation-regulated phases denote improving exposure duration.

At the same values of γ , the graph of the measured temperature (T) dynamics was a straight line: up to 15 min for manganese ions, 180 min for nitrite ions. The kinetics of extraction were limited by external mass transfer. A gradual decrease in the coefficient of external mass transfer indicated the deviation of the experimental points from the straight line characteristic of external diffusion, as the internal mass transfer begins to control the kinetics of the process. When comparing the modeling results, the most accurate description was given by the pseudo-II order model (coefficient of determination $R^2 = 0.983$), which suggests that the extraction had a mixed mechanism.

Analysis of the output curve of the purification degree of the sulfate ions model solutions (Figure 15) allows us to conclude that this sorbent is not effective for wastewater treatment from sulfate ions. In total for 5 cycles, the degree of purification was only 17.4%. When removing sulfate ions from an individual solution, the degree of purification under dynamic conditions for 1 cycle reached equilibrium; in the 2nd cycle, it was 4.9%. In total for 5 cycles, the degree of purification was 8.3% of sulfate ions from the monosolution.

The analysis of the output curves showed that the degree of purification of manganese from the model mixture (Figure 12) was 45% in the first cycle and decreased to 5% in the 2nd cycle; the final degree of purification was 35.4%. In the case of removing manganese ions from monosolution, the degree of purification under dynamic conditions decreased, so already in the second cycle, the decrease was from 55% to 15% from 1 to 5 cycles, respectively. The final degree of purification was 62.7%.

Experimental data showed that the total degree of purification from iron monosolution in 5 cycles (Figure 13) was 70.3%, from the model wastewater mixture, 98.1%. In the case of adsorption from a model mixture and a monosolution, the degree of purification decreased with each new cycle.

The degree of purification of nitrite ions on purolate (Figure 14) was 68.4% from the model mixture and 32.0% from the monosolution, and tended to decrease with each cycle. The results of the dynamics of nitrite ion extraction from the model wastewater solution showed that by the end of cycle 5, the total degree of purification reached 92.4%, and in the case of monosolution, 29.4%.

Some works [42–50] studied the properties of anthracite-based sorbents. The study [42] shows that traditional methods of mine water purification are expensive. Adsorption is the most cost-effective water purification method and is used as an inexpensive, efficient, and environmentally friendly method of purification of mine waters and process water. Approximately 80% of the ore is waste from metal mining. The development of sorbents from solid waste, including anthracite, is one of the promising solutions for solid waste management. The main provisions of the theory of adsorption of anions and cations on anthracite-based sorbents are considered.

In the article [44], the aspects of integrated wastewater treatment technology and sludge disposal of wastewater treatment plants are investigated. The authors substantiate the possibility of wastewater purification from copper and manganese ions using a sorbent based on purolate. The adsorption capacity of anthracite from the Sokyrnitsky deposit in relation to copper and manganese ions, depending on its fractional composition, is studied. Comparative studies of the efficiency of the adsorption of metal ions by natural zeolite and anthracite were used. The optimal sorption conditions for these ions were determined based on the findings of a study into the kinetics of their adsorption from Cu^{2+} and Mn^{2+} solutions [43]. While these studies underscore the potential within anthracite-based sorbents, they center mainly upon economic viability, together with basic adsorption theory, and fail to connect structural attributes to overall performance. Our findings augment this research by describing how ion elimination efficacy is regulated. Crystallographic ordering coupled with trace-element composition was ascertained via XRD and XRF analysis.

The study [44] reports on the findings of rapid sand filtration, a commonly used method for removing Fe, Mn, and NH_4 from oxygen-depleted groundwater used in drinking water supplies. This research combines microbiological and geochemical information to examine how the age of a filter composed of anthracite overlying quartz sand, which was in use for approximately 2 months to approximately 11 years, influences the sorption of Fe, Mn, and NH_4 . The removal of Mn and Fe from monosolutions was established, which was reflected in the coatings of the filter material. At the same time, ferrihydrite was formed in anthracite in the upper part of the filters (<1 m), and Mn oxides of the birnessite type were formed mainly in sand (>1 m). The Mn removal efficiency was increased by up to 28% within 2–3 months after the filter replacement and reached 100% after 8 months. However, after 11 months, the Mn removal decreased to between 60% and 80%. This lack of Mn removal with the new filters at 2–3 months may have been due to a relatively small amount of mineral coatings, which adsorbed Mn^{2+} ions and created surfaces that supported microbial growth. The removal of Fe^{2+} ($>90\%$) mainly occurred during the first 3 months of operation. This work provides valuable insight into the role of filter age and mineral coatings in Fe and Mn removal, but it does not characterize the sorbent material itself. By contrast, our study directly connects the physicochemical properties of purolate to its adsorption capacity, offering a material-level explanation for performance differences.

He et al. [45] indicated that nitrite ions are among the main pollutants of drinking water, and granular anthracite is frequently used as a filtration material in water treatment processes. In this study, the behavior of nitrite ions onto anthracite at adsorption under different temperature conditions was investigated using kinetic adsorption techniques, isotherm modeling, and the center of energy distribution theory. Nitrite adsorption onto anthracite is an endothermic process. Pseudo-second-order kinetics and Langmuir-Freundlich isotherm models can adequately describe the adsorption. The adsorption capacity of the material was $402.51 \text{ mg NO}_2\text{-N kg}^{-1}$ at 298 K, and it could be increased to $1,380.1 \text{ mg NO}_2\text{-N kg}^{-1}$ when the temperature reached 308 K. In addition to this, initially high-energy adsorption sites were occupied by nitrite ions, which subsequently diffused to lower-energy adsorption sites on anthracite. Furthermore, at higher temperatures, as the adsorption capacity improved, the thickness of the boundary layer increased, and the nitrite ions were predominantly absorbed through chemical means. Additionally, the adsorption process is facilitated by neutral pH. The adsorption may be limited by the presence of coexisting ions. Finally, using a 0.2 mol/l HCl solution, saturated anthracite can be effectively regenerated. Therefore, granulated anthracite is used as a filtering material and can also be used as a sorbent for the removal of nitrite ions. But in contrast to the data obtained by He et al. [45], our study adds novelty by showing how co-existing ions and trace impurities influence adsorption selectivity, which their work did not address.

The study [46] considered environmental pollution resulting from human activities and the most effective methods for protecting the biosphere using carbon-adsorption technologies. Due to their unique physical and chemical characteristics, AC materials represent an ideal adsorbent, allowing for the resolution of a wide range of environmental challenges.

The Dubinin assumption [46] allowed confirming that anthracite represents an excellent raw material to produce AC. Technology for producing Crushed Anthracite Sorbent (CANS) AC was developed. CASE AC was tested

in applications, such as drinking and wastewater purification, atmospheric gas and vapor protection, and soil detoxification of herbicide residues. The necessity of creating new opportunities to produce AC based on anthracite in the coal basins of Russia was substantiated.

In study [47], the influence of the type of raw material, the pre-carbonization process, and activation method on AC properties was investigated. Chemical (KOH and H_3PO_4) and physical (CO_2) activation was carried out on slowly pyrolyzed and hydrothermally carbonized biochar derived from two types of feedstocks (Scots pine, willow bark). In addition, the adsorption capacity of AC with two dyes and metallic zinc was tested. Clear differences were found between bio-chars and AC in terms of pore size distribution, surface area ($238\text{--}3505\text{ m}^2\text{ g}^{-1}$), and surface chemical composition. The activation of KOH led to the formation of highly microporous ACs from all bio-chars, whereas when using H_3PO_4 and CO_2 , an increase in the meso- and macroporosity of bio-chars was also observed. The adsorption capacity of the dyes depended on the surface area, and for zinc—on the pH of the AC [47]. Our study has some similarities with these works and complements them by focusing on natural anthracite-based purolate without chemical activation, thereby demonstrating its practical potential in real wastewater treatment contexts.

As shown in the study [48], adsorption is a highly effective process employed in wastewater treatment. This process is cost-effective, simple, environmentally friendly, and free from chemicals, making it an effective method for removing various types of inorganic and organic contaminants from different water sources. Due to these factors, it is widely utilized for this purpose. In addition, a minimal amount of residue is formed and there are no unwanted side products. The adsorbent can also be reused. This reduces operating expenses. The adsorption process is not only utilized independently as a standalone purification technique, but it is also frequently employed as a subsequent treatment to enhance the removal of contaminants from wastewater that have not been removed by oxidation. Researchers have suggested that the removal of ions from solutions occurs due to the bonds between ions and the surface functional groups of purolate [37–39, 49, 50]. Nevertheless, material-specific functioning is frequently disregarded in actuality. We diminish this disparity via correlating Fe, Mn, SO_4^{2-} , and NO_2^- removal efficiencies experimentally quantified within our purolate structural features study.

Boruk et al. [51] studied and discussed the effectiveness of anthracite treatment of wastewater generated during the production of sunflower oil in comparison with other water treatment technologies. Relatively small amounts of anthracite (10%–15% by weight of wastewater) provide effective disinfection and removal of up to 70% of pollutants in wastewater. This approach does not require major changes in production technology and can be easily implemented at existing production facilities. This effect is based on the double structure of the anthracite surface, consisting of hydrophilic and hydrophobic areas. The first ones provide high humidity of the adsorbent, as well as release some of the inorganic ions and provide coagulation of emulsified water. The latter adsorb and remove dissolved and portable organic pollutants from wastewater. After extraction, the spent sorbent can be cleaned and disposed of in an environmentally friendly way by burning coal as an additive to traditional fuels.

The review by Simate et al. [52] investigates the use of anthracite as an inexpensive adsorbent in wastewater treatment in various industries. The researchers analyzed the chemical composition and sorption properties of anthracite. The study indicates that, although the adsorption capacity for pollutants in various types of coal-based adsorbents may be significantly lower compared to other adsorbent materials, the significantly lower cost associated with coal offers significant potential for utilizing coal as a method for water treatment and the removal of various pollutants from wastewater. We agree on this point and expand it by providing the evidence that even low-cost anthracite-based carbons can achieve competitive performance when properly applied under multi-ion conditions.

Based on the literature review and market data, we evaluated the estimated costs of anthracite-based sorbents against conventional adsorbents. This analysis is not intended for the provision of an all-inclusive economic model. The intention was instead to feature purolate's corresponding financial benefit when scaled. Regarding water treatment, Table 12 summarizes typical cost ranges reported, accompanied by notes, for raw anthracite, natural zeolite, crushed anthracite sorbent, and commercial AC.

Table 12. Comparative cost estimates of sorbent materials

Sorbent Material	Typical Cost (USD per ton)	Notes on Use in Water Treatment
Raw anthracite (Kuzbass)	80–150	Abundant, minimal processing, used directly as filter media or after mild treatment
Crushed anthracite sorbent (CANS)	150–250	Produced from coal preparation plants; tested in wastewater and gas purification
Commercial AC	1,200–2,500	High adsorption capacity, but costly due to the activation process
Natural zeolite	200–400	Good ion-exchange capacity, but availability depends on the region

Source: Compiled by authors based on [46, 51, 52]

As summarized in Table 11, raw and processed anthracite materials are at least one order of magnitude less expensive than ACs. This economic advantage, combined with the demonstrated selectivity toward Fe and Mn, underpins the practical scalability of purolate-based sorbents for industrial wastewater treatment.

4 Conclusions

The physicochemical properties of carbon materials based on anthracite (purolate) were studied. In addition to carbon and oxygen, Zn (5,346.8 mg/kg), Ba (256 mg/kg), Sr (304 mg/kg), Cu (541 mg/kg), and MnO (119 mg/kg) were present in significant amounts in the purolate samples. The samples did not contain elements such as Al_2O_3 , SiO_2 , Rb, or Zr. When comparing the physicochemical properties of samples of different AC sorbents based on experimental and published data, we found that purolate had the largest particle size range (from 0.1 to 3 mm). Organic sorbents, compared to purolate, had pores ranging from 0.5 to 1.7 mm. However, the total volume of pores in water in purolate samples was the smallest $-0.5 \text{ cm}^3/\text{g}$. Unlike the structure of other AC sorbents, there were no mesopores in the structure of purolate. The data obtained during the analysis indicate that purolate has the largest pH range of aqueous extract, 8–9 units.

In the course of the study using the Rietveld method in the Topas (Bruker) program, the phase content was determined using the modeling of diffraction patterns. However, it was found that the modeling results were within the limits of errors of the XRD method. The data from the analysis of the adsorption breakthrough curve indicate that the service life of the sorbent layer was 380 min at an adsorption temperature of 28–30 °C.

When studying the adsorption of ions from aqueous solutions using purolate, for the isolation of manganese ions from aqueous solutions, the highest adsorption value (0.3 g/g) was observed for $C_{\text{equ}} 3.5 \text{ g/dm}^3$. When iron ions, nitrite ions, and sulfate ions were isolated, the highest adsorption was 0.14 g/g with $C_{\text{equ}} 4.0 \text{ g/dm}^3$, 0.1 g/g, with $C_{\text{equ}} 2.0 \text{ g/dm}^3$, and 1.2 g/g with $C_{\text{equ}} 6.0 \text{ g/dm}^3$, respectively. Analysis of the adsorption curves showed that the degree of removal of manganese from the samples of the model mixture was 45% in the first cycle and decreased to 5% in the 2nd cycle, and the final degree of purification was 35.4%. In the case of removal of manganese ions from monosolution samples, the degree of purification under dynamic conditions decreased. The decrease was from 55% to 15% from 1 to 5 cycles, respectively. The final degree of purification was 62.7%. Experimental data showed that the total degree of purification in 5 cycles from iron monosolution samples was 70.3% and from samples of the model wastewater mixture, 98.1%. In the case of adsorption from samples of model mixtures and monosolutions of manganese, iron, nitrites, and sulfates, the degree of purification decreased with each new cycle. The degree of purification from nitrite ions using purolate was 68.4% for samples of the model mixture and 32.0% for samples of monosolution, and tended to decrease with each cycle. The results of the dynamics of nitrate ion extraction from samples of the model wastewater solution showed that by the end of cycle 5, the total degree of purification reached 92.4%, and in the case of monosolution, 29.4%. Experimental data showed that the use of a purolate sorbent allows to effectively purify solutions from both cations and anions. Based on the analysis of kinetic curves of ion extraction, we concluded that the highest adsorption (0.07 g/g) for 250 min was observed during the extraction of manganese ions, and the lowest (0.045 g/g) for 300 min, during the extraction of nitrite ions.

Despite the performance demonstrated, several limitations must be acknowledged. Initially, the experiments were performed within controlled laboratory conditions with defined single-ion and model multi-ion systems. Authentic industrial discharges frequently include a rather detailed amalgamation of both organic and inorganic pollutants that might impinge upon adsorption. Subsequently, enduring constancy coupled with revitalization of purolate may fluctuate according to effluent makeup, given the noted impact from trace elements (e.g., Zn, Ba, Sr) upon sorption selectivity. Continued functionality, pressure attenuations, and operational duration throughout sustained activity require additional scrutiny to complete filters. Subsequent studies must consider these aspects to corroborate that purolate-based sorbents are applicable for complex industrial wastewater treatment.

Sorbents play a huge role in the oil and gas industry and ecology, where the study of the processes of sorption of substances is only gaining momentum. Dozens of patents are registered annually for inventions of new ways to separate some extracted minerals from others and improve traditional methods. These facts indicate that the study of sorption processes is at the peak of its development. It is necessary to study these processes using various sorbents in different conditions.

Author Contributions

Conceptualization: E.M.; methodology: E.U. and S.S.; formal analysis: E.U., S.S., S.I., and E.M.; writing—original draft: E.U. and S.S.; project administration: E.M.; writing—review & editing: S.I. and E.N.; supervision: E.N. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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